

Data Science

A stylized illustration in the background features a man and a woman interacting with various data visualizations. The man, seen from behind, stands near a large donut chart with segments in red, blue, and yellow, and a central circular inset showing a document icon. The woman stands to the right, pointing at a bar chart with red bars. In the foreground, a red bell curve is overlaid with vertical dashed lines and numerical labels: 20, 50, 65, 45, and 30. Other elements include a cloud with a list icon, a magnifying glass, and a control panel with colored buttons and sliders.

Lab Session 1

Lab Steps



Step 1 - Form groups
and sit in your groups
(3 members)



Step 2 - Send me the
list of your members



Step 3 - Have
Google Colab up and
running



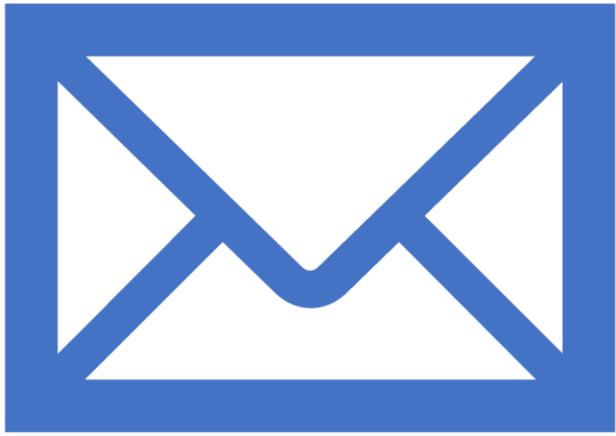
Step 4 - Have you
watched this
week's materials



Step 5 - Lab
assignment for today



Form groups and sit
in your groups (3
members)



Send me the list of
your members!

Have Google Colab up and running



Have you
looked at
this week's
materials?

- Webscraping
 - Video:
 - [Intro to Beautiful Soup - Tutorial](#)
 - [Selenium by Mariama](#)
 - [Selenium by Motunrayo](#)
 - Code:
 - [Intro to Beautiful Soup](#)
 - [Selenium Tutorial](#)

Lab assignment

Step1- Webscraping

- Choose a city in the world on the condition that there are at least 50 accommodations to book in its neighborhoods.
- Choose a date for the trip
- Scrape the data about the available accommodations for that city from three accommodation booking websites, e.g., Booking.com, Hotels.com, Expedia, Agoda, etc.
- What are the challenges in scraping the data? How do you store the scraped data for a further easy use?

Step 2 – Data pre-processing

- What are the data processing steps for having a proper dataset to be used in the next step?

Step 3 – Exploratory data analysis (EDA)

Use the appropriate visualisation method to provide a comprehensive EDA, e.g., price ranges, distance from city center, number of starts, review score, etc. It is up to you what features to choose to visualise.

Step1 - Webscraping



Finding the solution - 20 min



Team presentation - 10 min

Step 2 - Data Preprocessing



Finding the solution - 20 min



Team presentation - 10 min

Step 3 - EDA



Finding the solution - 20 min



Team presentation - 10 min



What is next?

Step 4 - Interaction with the user



Step 4 – Interaction with the user

Your code should be able to perform the following task to give the user the list of “perfect” accommodation in the city:

- User inputs certain preferences for the accommodation, , e.g., price ranges, distance from city center, number of stars, review score, etc.
- User chooses the criteria that based on which the accommodations should be sorted
- The code returns the sorted list of all accommodations in the area
- The user should be able to input the name of a hotel from your list and get the following information:
 - Hotels address
 - Distance from the city center
 - Average price per night for given accommodation type for the data that you have pre-selected
 - Review score
 - Hotel rating
 - Hotel summary/description
 - At least 3 photos of the hotel